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NSMA: A Hybrid NSGA-II and Local Search Approach for Multi-Objective Railway Hub Location and Network Design

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Abstract. The Hub Location Problem (HLP) is a cornerstone of transportation network optimization, particularly for railway systems where cost efficiency and service quality are critical. This study formulates railway infrastructure planning as a multi-objective single allocation Hub Location and Network Design (HLND) problem, addressing trade-offs between system cost, service quality, and user experience under realistic constraints such as incomplete hub networks. We propose NSMA, a novel hybrid algorithm combining the Non-dominated Sorting Genetic Algorithm II (NSGA-II) with local search, to efficiently solve the HLND problem. Applied to a dataset of 54 train stations along Thailand's Northern Line from Bangkok to Chiang Mai, NSMA outperforms the NSGA-II baseline on key metrics, including hypervolume and inverted generational distance, demonstrating superior solution quality and convergence. This approach generates practical Pareto-optimal solutions, offering decision-makers valuable insights for sustainable railway network design. This work advances HLP applications by integrating advanced metaheuristics with region-specific railway planning, addressing gaps in multi-objective optimization for large-scale rail systems.

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Key words: Hub location, NSGA-II, Railway transport, Multi-objective Optimization

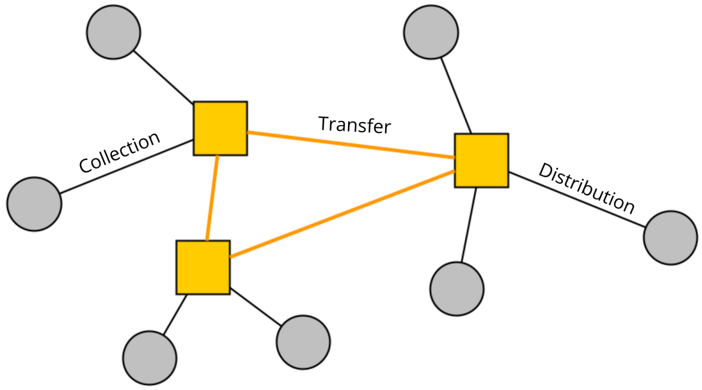

Figure 1 **Hub and Spoke network**

1. Introduction

Hub Location Problem (HLP) is an outstanding problem in location science, not only for its wide range of applications, but also for its challenges. Originating from the facility location problem, it serves as a critical framework in operations research, focusing on the optimal placement and operation of hubs within a network to facilitate efficient flow of goods, services, or passengers. HLP has found widespread applications across various sectors, including air transportation (Bryan and O'Kelly 1999), freight logistics (Alumur and Kara 2008), postal services (Çetiner, Sepil, and Süral 2010), and maritime transport (Konings 2006).

The Hub and Spoke network model underpins many HLP applications. This model leverages economies of scale, enabling cost reductions by consolidating large flows of goods or passengers through dedicated routes, often at a discounted rate α. In this system (Figure 1), demands from multiple origins are aggregated at central hubs, transferred via high-capacity, specialized lines, and then distributed to their final destinations. This structure minimizes the number of direct connections, reduces transportation costs, and enhances network efficiency, making it particularly suitable for large-scale systems.

Railway transport plays a vital role in global transportation systems, driven by a growing emphasis on sustainable and public transportation solutions. With increasing concerns about carbon emissions and urban congestion, governments and organizations worldwide are shifting toward greener alternatives, such as rail transit, to meet environmental and societal demands (Banister 2007). HLP models have been adapted to address challenges in railway systems, such as optimizing station placement and network design to enhance connectivity and efficiency (Jeong, Lee, and Bookbinder 2007). By treating train stations as hubs and railway lines as spokes, HLP provides a robust framework for improving rail infrastructure planning and operations.

Despite the progress in applying HLP to railway systems, significant research gaps remain. Many existing HLP models for rail transportation are either overly simplistic or impractical, often failing to account for real-world constraints such as multi-objective trade-offs, infrastructure limitations, and varying demand patterns (Alumur et al. 2021). Furthermore, there is a lack of advanced optimization techniques tailored specifically for railway hub location problems. This gap hinders the development of efficient and practical solutions for large-scale rail networks, necessitating innovative approaches to address these challenges effectively.

The HLP is classified as an NP-hard problem (Kara and Tansel 2000), meaning that it is an intractable problem, i.e. a problem that finding optimal solutions for large instances is computationally intensive and infeasible within practical timeframes. (Ernst et al. 2009) prove that the single allocation subproblem with respect to a given set of hubs is already NP-hard. Exact methods, such as branch-and-bound, guarantee optimality but are limited by their scalability (Ernst and Krishnamoorthy 1996). As a result, researchers have increasingly turned to heuristic and metaheuristic approaches to find near-optimal solutions efficiently (Kratica et al. 2007). Developing sophisticated metaheuristics is essential to balance solution quality and computational efficiency, particularly for complex, real-world railway network design problems.

This study makes several key contributions to the field of railway infrastructure planning. First, we formulate the railway infrastructure planning problem as a multi-objective single allocation hub location and network design (HLND) problem, incorporating service-based objectives and user experience besides system cost. Second, we propose a novel hybrid algorithm, named NSMA, which integrates the Non-dominated Sorting Genetic Algorithm II (NSGA-II) with Local Search to address the HLND problem effectively. The algorithm is applied to a dataset comprising 54 train stations along the Northern Line, from Bangkok to Chiang Mai, Thailand, providing decision-makers with valuable trade-off insights for optimizing railway network design.

The rest of this paper is organized as follows: Section 2 reviews related works on hub location problems and their applications in transportation. Section 3 presents the mathematical formulation of the proposed model. Section 4 describes the proposed NSMA algorithm for solving the model and its components. Section 5 discusses the experimental results of the algorithms applied on the case study dataset. Finally, Section 6 concludes the paper and outlines future research directions.

2. Related Works

Hub Location Problem (HLP) has evolved significantly since its inception in the late 1980s. The work by O'kelly (1987) had lay the foundation for the field by migrating theory from facility location. He introduced the very first formulation for HLP, which is a quadratic integer programming formulation. (Campbell 1994) provided integer programming formulations for discrete hub location problems, introducing the p-hub center and hub covering problem. (Ernst and Krishnamoorthy 1996) formulated the uncapacitated single allocation HLP as a multi-commodity flow problem with an efficient model of size $O(n^3)$. For more insights on the HLPs, please refer to the surveys by (Alumur and Kara 2008), Campbell and O'Kelly (2012) and Alumur et al. (2021).

A key extension of traditional HLP is the consideration of incomplete hub networks, where the hub network is not necessarily fully inter-connected. In practice, costs for building and maintaining hubs facilities and network infrastructure is way higher than transportation cost. Thus, establishing a full hub network is not realistic. Alumur, Kara, and Karasan (2009) proposed formulations for single allocation incomplete hub networks. de Sá, Morabito, and de Camargo (2018) applied Benders decomposition to solve a multiple allocation incomplete hub location problem with uncertain demands.

Multi-objective optimization is often used to modeling real-world problem, where tradeoffs must be made between conflicting objectives. According to (Alumur et al. 2021), there are some main types of objectives in HLP: economic-based, such as cost (median problem) or profit; and service-based objectives, like center and covering problems. Rahimi et al. (2016) developed a multi-objective hub network design under uncertainty, using an M/M/c/K queue system to model congestion at hubs, optimizing for cost and service levels. In cold supply chain, Golestani et al. (2021) investigates a multi-objective green hub location problem. From the Mixed-integer linear programming (MILP), they used the ϵ-constraint method to obtain the Pareto frontier. To solve these complex problems, Multi-Objective Evolutionary Algorithms (MOEAs) are favored due to their population-based nature and the ability to generate near-optimal Pareto solutions. Non-dominated sorting genetic algorithm-II (NSGA-II), is one of the most well-known MOEAs widely adopted in many fields (Deb et al. 2002). Demır, Ergın, and Kıraz (2019) used the NSGA-II algorithm to solve the multi-objective capacitated multiple allocation hub location problem (MOCMAHLP). NSGA-II is also successfully applied in (Eghbali, Abedzadeh, and Setak 2014) for a multi-objective reliable hub covering problem.

In the context of railway transportation, HLP applications address unique challenges such as fixed infrastructure and high-volume freight/passenger flows. Jeong, Lee, and Bookbinder (2007) modeled the European freight railway system as a hub-and-spoke network. More recent works incorporate practical considerations, including capacity constraints and multi-modality. Chanta and Sangsawang (2021) used a two-stage approach for finding optimal location of station in a rail transportation network. Chanta et al. (2024) studied a bi-objective hub location model considering congestion for railway transportation planning. Arnold, Peeters, and Thomas (2004) deals with the problem of finding optimal location for rail/road freight transport terminals.

Despite these advances, gaps remain in addressing multi-objective optimization for railway hub location problems with realistic constraints, such as single allocation and incomplete hub networks. This work contributes by formulating railway infrastructure planning as a multi-objective single allocation Hub Location and Network Design (HLND) problem, capturing trade-offs between cost and service quality. We propose the NSMA algorithm, a hybrid of NSGA-II and local search, to improve solution quality for large-scale instances. Applied to a dataset of 54 train stations along Thailand's Northern Line, our approach provides actionable trade-off insights for decision-makers, building on existing methods with a focus on practical, region-specific railway planning.

3. Mathematical Formulation

In order to rigorously define the problem, one needs to establish a comprehensive mathematical formulation. The formulation must capture the details of the real-world problem being modeled. In a Mixed Integer Programming (MIP) model, the following components need to be included:

- Input parameters
- Assumptions of the model
- Decision variables
- Objectives for optimization
- Constraints

3.1. Input parameters

We denote:

- N set of nodes, indexed by $1, 2, 3, \ldots, n$
- $H \subseteq N$ set of potential hub nodes, indexed $1, 2, \ldots, h$
- w demand (flow) matrix, w_{ij} is the amount of flow to be sent from origin node $i \in N$ to destination node $j \in N$. In this context, w_{ij} is the amount of people want to go from i to j.
- C_{ij} transportation cost from node $i \in N$ to node $j \in N$.
- t_{ij} : time to travel from $i \in N$ to $j \in N$.
- FH_k fixed cost for opening a hub with unlimited capacity at node k $(k \in H)$
- FL_{ij} fixed cost to open the hub link (i, j) (with $i, j \in H$)
- t_k : the delay time at hub node $k \in H$. This was suggested in Chanta et al. (2024) to account for congestion at train stations.

There're also some constants:

- $\alpha (0 < \alpha < 1)$ discount factor for transportation cost on hub links for economies of scales
- $\beta (0 < \beta < 1)$ discount factor for traveling time on hub network
- χ collection coefficient (for transportation cost from origin node to hub node)
- δ distribution coefficient (for transportation cost from hub node to destination node)
- L : constant distance defining demand coverage. Any node allocated to a hub within this distance L is considered covered.

For convenience, define:

- $O_i = \sum_{j \in N} w_{ij}$ total flow originating from node i
- $D_i = \sum_{j \in N} w_{ji}$ total flow destined at node i

3.2. Model assumptions

1. **Single allocation**: Each node is allocated to exactly one hub.
2. Direct connection between non-hub nodes is disallowed, therefore, the path from node i to node j must go through at least one hub node.
3. **Connectivity**: The hubs network are *strongly* connected, i.e., for every hub node i there exists a directed path to every other hub node j via the arcs in the hub network.
4. Hub links are uni-directional. Thus, cost matrix C does not guarantee symmetry, i.e., $C_{ij} \neq C_{ji}$ for some $i, j \in N$.
5. Cost matrix C satisfies triangle inequality, i.e. $C_{ij} \leq C_{ik} + C_{kj} \quad \forall i, j, k \in N$
6. **Uncapacitated**: There is no upper limit on the amount of flow that can pass through a hub node or link.
7. The model is deterministic. All inputs are known with certainty.

3.3. Decision variables

Decision variables can be seen as the output, or solution that the model needs to find. In this model, we define the following decision variables:

- $x_{ij} = 1$ if node $i \in N$ is allocated to the hub node $j \in H$, 0 otherwise
- $x_{kk} = 1$ if node k is a hub.
- $z_{ij} = 1$ if the hub arc from i to j is established $(i, j \in H)$, 0 otherwise;
- f_{ij}^k is the total amount of flow (commodity) originating from node $k \in N$ and routed on hub arc (i, j) in that order (from i to j)

To formulate the p-hub center and covering problem on incomplete hub network, Alumur, Kara, and Karasan (2009) used the following variables:

- $y_{ij}^k = 1$ if the spanning tree rooted at hub $k \in H$ uses the hub arc (i, j); else 0
- s_{ij} the undiscounted travel time from hub $i \in H$ to hub $j \in H$ in the hub network, including the total delay time at hub nodes on the path.

For convenience, let us denote $X_i = \sum_{k \in H} k x_{ik}$ index of the hub node allocated to i. X concides with the *allocation vector*, which we will see in Section 4

3.4. Objectives

The model has to simultaneously optimize three conflicting objectives as below:

$$\text{Minimize} \quad Z_1 = \sum_{i \in H} \sum_{j \in H} \mathrm{FL}_{ij} z_{ij} + \sum_{k \in H} \mathrm{FH}_k x_{kk} + \sum_{i \in N} (\chi C_{ik} O_i + \delta C_{ki} D_i) x_{iX_i} + \sum_{i \in H} \sum_{j \in H} \sum_{k \in N} \alpha C_{ij} f_{ij}^k$$

$$\text{Minimize} \quad Z_2 = \max_{i,j \in N} \left\{ t_{iX_i} + \beta s_{X_i X_j} + t_{X_j j} \right\}$$

$$\text{Maximize} \quad Z_3 = \sum_{i \in N} O_i [C_{iX_i} \leq L]$$

Note that $[B] = 1$ if B is true, 0 otherwise. Expression $[B]$ is represented by a binary variable.

Z_1 is the total cost, including fixed costs and transportation costs. The first term contains fixed costs for building hubs and establishing hub links. The second term is the total transportation cost including (a) cost on the collection leg (from origin nodes to hub nodes, multiplied by χ), (b) cost on the distribution leg (from hub nodes to destination nodes, multiplied by δ), and (c) transportation cost on the hub network, multiplied by discount factor α to account for economies of scale.

Z_2 is the maximum travel time across all origin-destination pairs discounted by β. The travel time comprise of time to move from origin i to its hub X_i, from origin hub X_i to destination hub X_j via hub network, and from destination hub X_j to destination node j.

Z_3 is the total demand covered by the hub network within the distance L. We hypothesized that if a passenger is allocated to a train station too far away from their origin, they may not use the service. Thus, Z_3 serves as an indicator for the quality of service and accounts for user experience and accessibility.

With one cost-based objective (Z_1) and two service-based objectives (Z_2 and Z_3), the model aims to find a balance between minimizing costs and maximizing service quality, combining the perspectives of both service providers and users in the optimization process.

3.5. Constraints

The model is subjected to the following constraints:

$$\sum_{j \in H} x_{ij} = 1 \quad \forall i \in N \tag{1}$$

$$x_{ik} \leq x_{kk} \quad \forall i \in N, k \in H \tag{2}$$

$$z_{ij} \leq x_{ii} + x_{jj} - 1 \quad \forall i, j \in H \tag{3}$$

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$$\sum_{j\in H:j\neq i} f_{ji}^k + O_k x_{ki} = \sum_{j\in H:j\neq i} f_{ij}^k + \sum_{l\in N} w_{kl} x_{li} \quad \forall i \in H, k \in N \tag{4}$$

$$f_{ij}^k \leq O_k z_{ij} \quad \forall i,j \in H, k \in N \tag{5}$$

Equation (1) ensures single allocation property. Eq. (2), (3) ensures valid allocation and establishing of arcs. Eq. (4) is the flow divergence equation, and Eq. (5) ensures flow can only pass through established arcs.

$$\sum_{i\in H, i\neq j} y_{ij}^k \geq x_{jj} + x_{kk} - 1 \quad \forall j,k \in H : j \neq k \tag{6}$$

$$\sum_{i\in H, i\neq j} y_{ij}^k \leq x_{kk} \quad \forall j,k \in H : j \neq k \tag{7}$$

$$y_{ij}^k \leq z_{ij} \quad \forall i,j,k \in H \tag{8}$$

$$y_{ij}^k + y_{ji}^k \leq 1 \quad \forall i,j,k \in H \tag{9}$$

$$s_{kj} \geq (s_{ki} + t_i + t_{ij}) y_{ij}^k \quad \forall i,j,k \in H : i \neq j \wedge j \neq k \tag{10}$$

$$s_{kk} = 0 \quad \forall k \in H \tag{11}$$

The constraints (6) – (11) construct a directed spanning tree rooted at each hub k based on z_{ij}, and are adapted from Alumur, Kara, and Karasan (2009). Then, the shortest distances are computed via s using edge-relaxation (10). Notice that delay time t_i is included in the computation of s_{kj}.

Finally, we have the following domain constraints:

$$f_{ij}^k \geq 0 \quad \forall i,j \in H, k \in N$$

$$x_{ij} \in \{0,1\} \quad \forall i,j \in N$$

$$z_{ij} \in \{0,1\} \quad \forall i,j \in H$$

$$y_{ij}^k \in \{0,1\} \quad \forall i,j,k \in H$$

$$s_{kj} \geq 0 \quad \forall j,k \in H$$

3.6. Characteristics of the model

- The mathematical formulation has $O(n^3)$ variables and constraints. There are $O(n^3)$ binary variables.
- All constraints are linear. The objective Z_2 can be linearized using ϵ-constraint method. Under single objective, the given formulation becomes a *Mixed Integer Programming* model and could be solved efficiently using standard optimization solvers.
- To handle multi-objective optimization, one may need to use several techniques, such as weighted sum method, ϵ-constraint method, or Pareto-based methods.
- This model is an adaptation of the work by Alumur, Kara, and Karasan (2009) and extension of the work by Chanta et al. (2024).

4. The proposed algorithm

The proposed algorithm NSMA is a multi-objective optimization algorithm developed from NSGA-II. In this section, we present a summary of NSGA-II, solution representation, components of the algorithm and algorithm flowchart.

4.1. The NSGA-II algorithm

NSGA-II, proposed by Deb et al. (2002), is one of the most well-known multi-objective evolutionary algorithms (MOEAs). It is an Elitist GA, where parents and offspring are combined and competing for survival at each generation. To enhance efficency, they developed a faster implementation for non-dominated sorting that quickly generates nondominated layers $F_1, F_2, \ldots, F_k$ in $O(MN^2)$ time comlexity. Population diversity is a crucial feature in evolutionary algorithms, especially for multi-objective optimization where multiple Pareto solutions are desired. Thus, the authors introduced crowding distance as a mechanism for promoting diversity. In the equation (4.1) x_{low}, x_{high} are the two points nearest to solution i in objective space m. $i_{distance} = \infty$ if one of its objective value is extreme.

$$i_{distance} = \sum_{m=1}^{M} \frac{x_{high}^m - x_{low}^m}{f_{max}^m - f_{min}^m}$$

The NSGA-II algorithm uses binary tournament selection, and an elitist selection process. In comparison of individuals, the one belong to front layers F_i would be prioritized over the one belongs to later, dominated layers $F_j : j > i$. For individuals belonging to the same layer, the one with the higher crowding distance would be preferred. The selection and replacement phases are re-used in our algorithm. However, solution representation, crossover and mutation operators are left to be designed for specific implementation for each problem.

4.2. Flowchart of the algorithm

Memetic algorithm is an extension of GAs that adds a Local Intensification phase at the end of every generation. In this phase, every individual undergoes the Local Search process. This technique helps improving the convergence and solution quality, and has been successfully applied in various works (see Neri and Cotta (2012)). NSMA is created by adding the Local Search phase to the NSGA-II framework (see Figure 2).

4.3. Solution representation

Solution representation plays an important role in meta-heuristic algorithms, as this choice will shape the whole algorithm designing process. In this work, the representation comprises of two

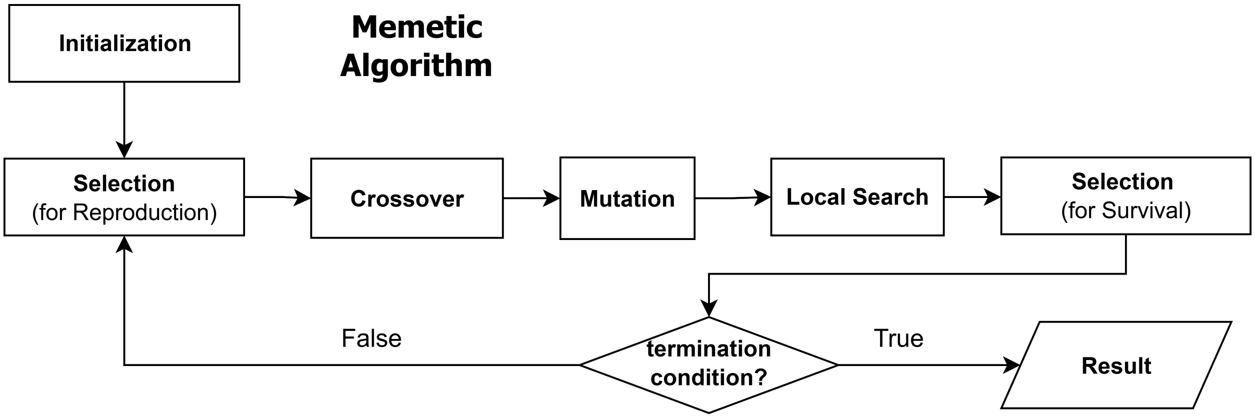

Figure 2 **Steps in a Memetic Algorithm**

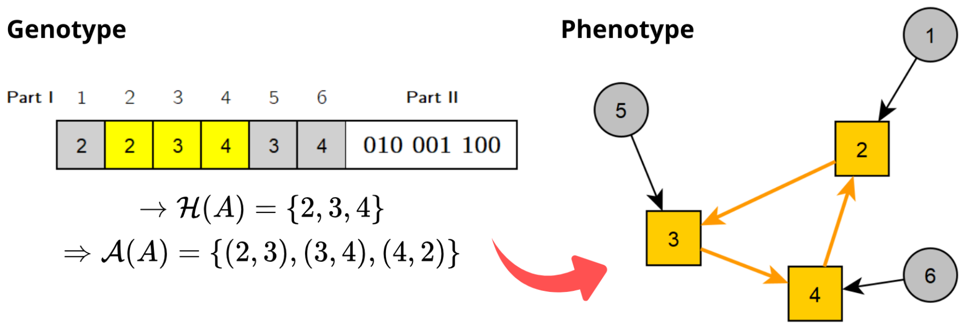

Figure 3 **Illustration of the solution representation, consisting of two parts**

parts. The first part contains information about hub location and allocation. It is an integer vector represent the allocation pattern of a solution, often refered in literature as the *allocation vector*.

- $A_i = k$ $(k \neq i)$ if node i is allocated to hub k; otherwise, $A_k = k$ if k is a hub
- The representation must satisfy constraint (2) (i.e. $A_{A_i} = A_i$)

For a representation A, denote $\mathcal{H}(A) = \{h_1, h_2, \ldots, h_p\}$ as the hubs that appear in A, then $p = |\mathcal{H}|$ is the number of hubs.

The second part of the solution representation contains hub network (arc location) information. It is a binary string of length p^2 where the $(i-1)p + j$-th bit corresponds to the directed arc (h_i, h_j).

Fitness value of an individual is a vector of three objective values (Z_1, Z_2, Z_3). The evaluation process goes as follows:

We assume the representation is valid, if not, apply the `repair()` method.

1. Compute all-pair shortest paths using algorithm Floyd (1962) in $O(n^3)$. The shortest travel time from i to j is stored in s_{ij}, and lowest transportation cost stored as c_{ij}. The computed distance satisfies triangle inequality, i.e. $s_{ij} + s_{jk} \geq s_{ik}$.
2. Loop over every O-D pair (i, j) and compute Z_1, Z_2, Z_3 in $O(n^2)$

3. Total time complexity for evaluation is $O(n^3)$

4.4. Crossover operator

The crossover operator is applied to each part of the solution representation. From a pair of parent (P_1, P_2), the crossover process produces a pair of offspring (C_1, C_2). The crossover operator has two phases corresponding to two parts of representation. The first phase is called "Hub crossover", and is developed from Uniform crossover and 1-point crossover techniques. The steps of Phase 1 are as follows:

1. **Hub exchange:** randomly distribute $\mathcal{H}(P_1) \cup \mathcal{H}(P_2)$ to $\mathcal{H}(C_1)$ and $\mathcal{H}(C_2)$ with uniform probability. The common hubs $\mathcal{H}(P_1) \cap \mathcal{H}(P_2)$ are inherited in both C_1, C_2. In essence, this corresponds to the Uniform crossover operator applied on the binary string representation of hub decision.

2. **Hub mapping:** create mappings $\mathcal{M}_{C_1}, \mathcal{M}_{C_2}$ for C_1, C_2 respectively. $\mathcal{M}_A(k) = k$ if $k \in \mathcal{H}(A)$, else $\mathcal{M}_A(h) = \arg\min_{k \in \mathcal{H}(A)} \text{distance}(k, h)$. If a hub h is not present in $\mathcal{H}$, then it is mapped to the nearest hub $k \in \mathcal{H}$.

3. **Crossover:** $C_1 \leftarrow P_1; C_2 \leftarrow P_2$, then 1-point crossover is applied on the allocation vectors C_1, C_2.

4. *Repair:* After crossover, C_1 and C_2 may contain hub not present in its hub set. Hub mappings $\mathcal{M}_{C_1}$ and $\mathcal{M}_{C_2}$ are used to fix C_1 and C_2's allocations, ensuring the validity of hub allocation.

As can be seen from figure 4, the two children share some traits with their parents.

The second phase is "Arcs crossover". Denote $\mathcal{A}(A)$ the arcs in the representation A. This phase consists of the following steps:

1. **Arc exchange:** randomly distribute $\mathcal{A}(P_1) \cup \mathcal{A}(P_2)$ to $\mathcal{A}(C_1)$ and $\mathcal{A}(C_2)$ with ratio $|\mathcal{H}(C_1)|^2/|\mathcal{H}(C_2)|^2$. The common arcs $\mathcal{A}(P_1) \cap \mathcal{A}(P_2)$ are inherited in both C_1, C_2.

2. **Arc mapping:** each arc (u, v) is mapped to $(\mathcal{M}_C(u), \mathcal{M}_C(v))$. Invalid self-loops and multiple edges are removed.

3. **Repair:** by fixing the strong connectivity property, using the `repair()` method.

4.5. Mutation operators

Mutation introduce slight modifications to the representation. Its purpose is to maintaining necessary genetic diversity and enable exploitation capability of GA. In this work, four mutation operators are used. The four mutation operators below has the same probability to be chosen ($\frac{1}{4}$):

- **m1 - Make hub**: select a random node i and make it a hub

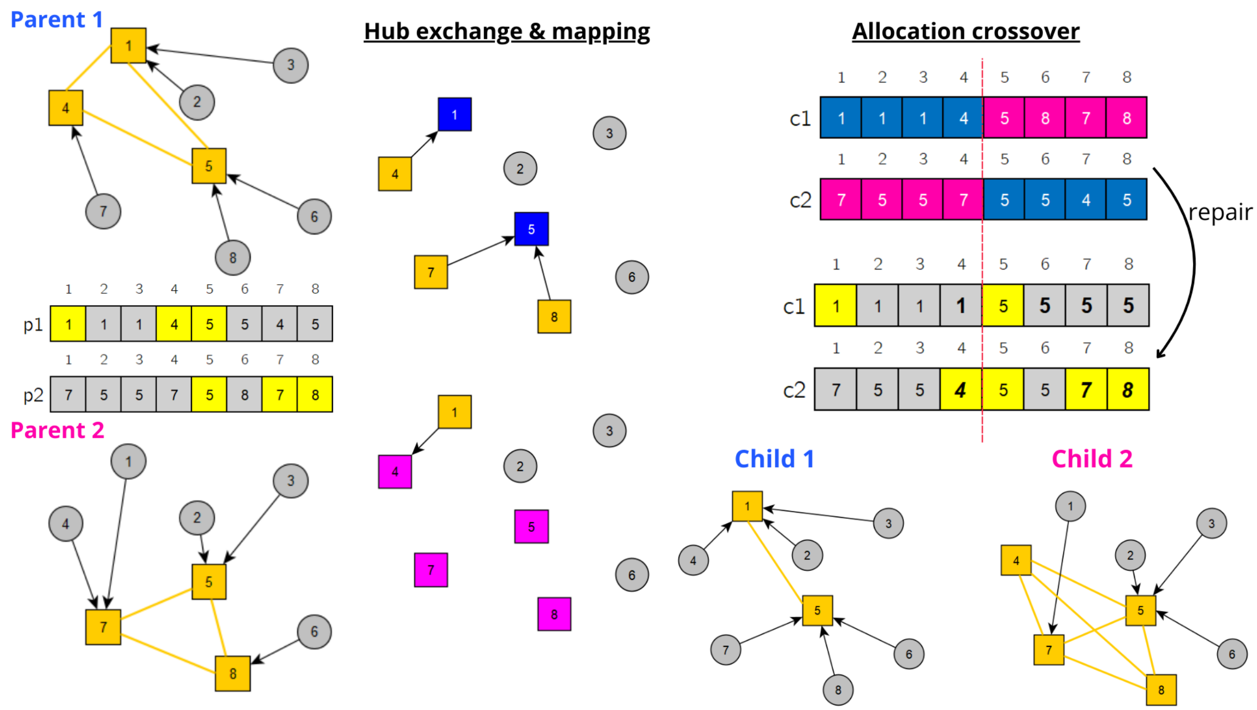

Figure 4 Illustration of the first phase of crossover operator

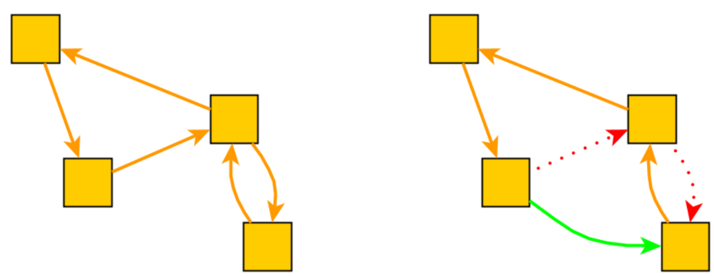

Figure 5 m4 flips 3 bits, adding 1 arc and removing 2 arcs

- **m2 - Erase hub**: select a random hub k and make it a node. All node j currently allocated to k is then assigned to a random hub (other than k).
- **m3 - Hub exchange**: exchange a hub k with a non-hub node i. All nodes currently assigned to k will be allocated to i.
- **m4 - Arc flip**: selects $\max\{2, \frac{|\mathcal{A}|}{p}\}$ random bits in the arc representation and flips them. If the flipped bit is 1, then the corresponding arc is removed, otherwise it is added.

4.6. Repair operator

The crossover operator and mutation operators oftenly break the strong connectivity property of the hub network. To fix this, we apply the `repair()` method. This method combines Kruskal's

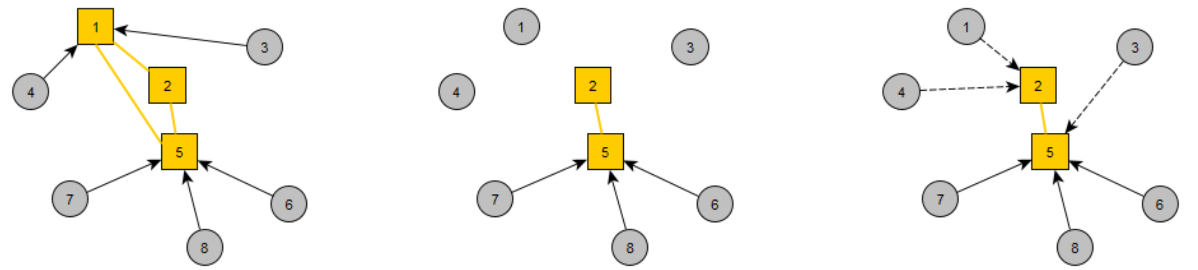

Figure 6 **m2 erases hub 1**

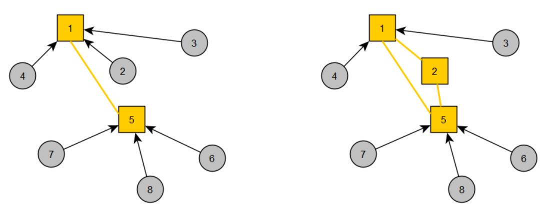

Figure 7 **m1 makes new hub 2**

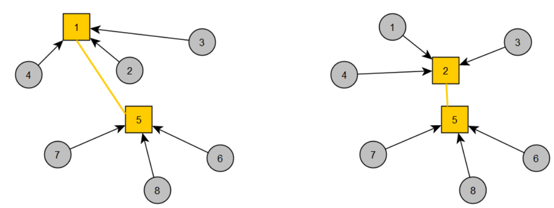

Figure 8 **m3 exchanges hub 1 with node 2**

greedy algorithm for finding minimum spanning tree with depth-first search to restore the strong connectivity property of the network. The method works as follows:

Denote $\bar{\mathcal{A}}$ the set of arcs **not** included in the representation A; and $a \rightsquigarrow b$ if node b is reachable from node a via the arcs $\mathcal{A}$

Algorithm 1 repair() method - fix strong connectivity

for $k \in H : A_k = k$ **do** ▷ for each hub k established
  while not $k \rightsquigarrow \mathcal{H} \setminus \{k\}$ **do** ▷ there is node un-reachable from k
    $(u, v) \leftarrow \arg\min_{i,j \in \mathcal{H}} \{C_{ij} \mid (i, j) \in \bar{\mathcal{A}} \mid k \rightsquigarrow i \wedge \neg k \rightsquigarrow j\}$ ▷ select the min-cost arc
    Add the arc (u, v) to the representation A
    $\bar{\mathcal{A}} \leftarrow \bar{\mathcal{A}} \setminus \{(u, v)\}$
    Perform Depth-first Search to update reachability status
  end while
end for

NSGA-II implementation for HLP

With the components defined, we now have a detailed implementation of NSGA-II for the hub location problem. The flowchart is at figure 9.

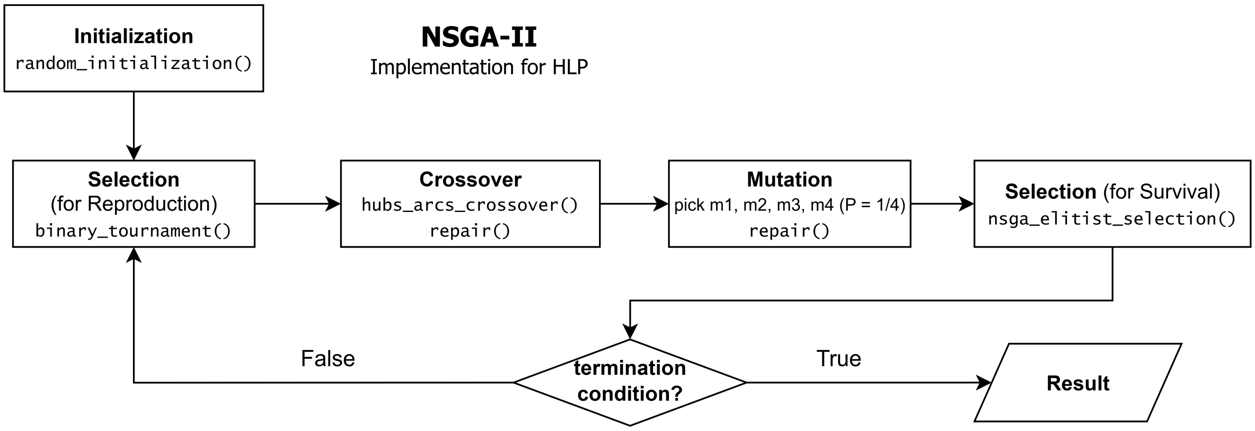

Figure 9 Detailed flow of NSGA-II algorithm for HLP

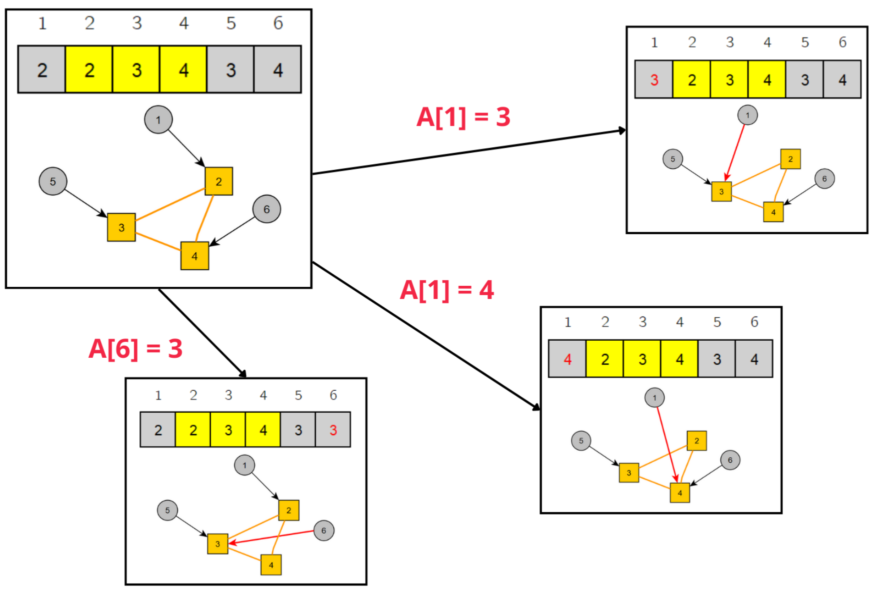

Figure 10 Neighborhood structure illustration

4.7. Local search

While the proposed operators can efficiently optimize the hub location and hub network structure, the allocation pattern is less frequently optimized. Local search is applied to optimize the allocation of nodes to hub given a fixed hub network and location.

In order to employ the local search technique, one must first define a neighborhood structure. We denote the neighborhood $\mathcal{N}(A)$ of A as the set of representation that differs from A at exactly 1 position (Figure 4.7).

$$\mathcal{N}(A) = \{A' \quad | \quad \text{distance(A,A')} = 1\}$$

A representation A has exactly $(n-p)(p-1)$ neighbors (where p is the number of hubs). The searching process loops through all neighbors iteratively and select the first one that dominates the

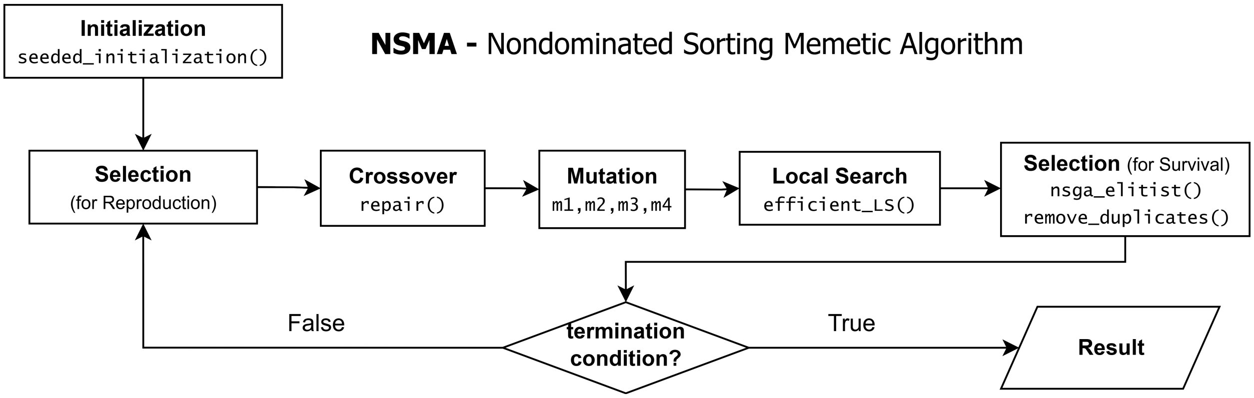

Figure 11 **Detailed steps of NSMA algorithm**

current solution. The process repeats until local optimum - i.e. no better neighbor is found. Local search is applied after every mutation and for every 25 generations.

However, for each neighbor, simply invoking `evaluate()` method will incur a time complexity of $O(n^3)$. We observe that changing an allocation only affect $O(n)$ components in the objective functions.

- $C_{iA_i}(\chi O_i + \delta D_i) + \sum_j \alpha c_{A_i A_j} w_{ij}$ (Z_1).
- $\{t_{iA_i} + \beta s_{A_i A_j} + t_{A_j j}\}$ (Z_2).
- $O_i[C_{iA_i} \leq L]$ (Z_3).

By adopting efficient data structures that account only for the incremental changes, we obtain an efficient implementation. Evaluation for each neighbor can be done in $O(n)$. The details of the implementation are as follows. First, compute incremental changes to Z_1, Z_3 by subtracting and adding corresponding components. Next, for Z_2, maintain $\mathcal{T}_i = \{\mathrm{tt}_{ij}\}$ - set of travel time to j for each i. Removes the i-th row and i-th column of the matrix $\mathcal{T}$, then compute max of remaining element, let it $M = \max_i T_i = \max_{i,j \neq i} t_{ij}$. For each new value of A_i, computes $\mathcal{T}_i{}'$, then $Z_2 = \max\{M, \max T_i'\}$. Using binary tree data structure, we can make max-value queries and single value updates on T_i in $O(\log n)$ time.

4.8. The flowchart of the proposed algorithm NSMA

The final algorithm is obtained, as shown in Figure 11. The algorithm starts with an initial population of random solutions. The population is then sorted into non-dominated layers, and the crowding distance is computed. The selection process is performed using binary tournament selection, where parents are chosen based on their non-dominated rank and crowding distance. The crossover and mutation operators are applied to generate offspring, which are then evaluated for their fitness values. The local search phase is applied to each offspring to further improve their allocation patterns. Finally, the population is updated by combining parents and offspring, selecting the best individuals based on non-domination rank and crowding distance. The process repeats until a pre-defined number of generations or convergence criteria is reached.

Parameter	NSGA-II	NSMA
POP_SIZE	500	500
NUM_GEN	200	200
p_c	0.95	0.95
p_m	0.3	0.24
$[f_{min}, f_{max}]$	$\{[5 \cdot 10^8, 10^{10}], [20, 800], [0, 93000]\}$	

Table 1 Algorithm parameters

5. Experimental results

We apply our proposed algorithm to a case study based on a dataset of train stations along the Northern Line in Thailand. NSMA algorithm is put into direct comparison with NSGA-II. Each algorithm is run for 20 times on each problem instance to have statistical significance. We use well-known evaluation metrics that are widely used for assessing multi-objective optimization algorithms. The mean value and standard deviation of these metrics are reported. Experiments are run on a laptop with CPU 11th Gen Intel(R) Core(TM) i5-1135G7 @ 2.40GHz with 16 GB RAM. Algorithm parameters are reported in Table 1.

As can be seen from Table 1, $[f_{min}, f_{max}]$ show that objective values have highly different scales. Therefore, they are transformed with the min-max normalization technique to ensure a uniform scale across all objectives. Each objective value now lies in the range $[0, 1]$

$$x' = \frac{x - \min}{\max - \min}$$

5.1. Dataset

The dataset is taken from Chanta et al. (2024), consists of 54 train stations along The Northern Line from Bangkok to Chiang Mai. The distance matrix is calculated using Haversine formula. The flow matrix contains the averaged amount of passenger from each origin station i to destination station j over a period of years.

There are two problem variants: **Complete** and **Incomplete**. From this dataset, two problem instances with size $n = 10$ (small) and $n = 54$ (medium) are derived for each problem variant. The parameter FL is only used in the Incomplete variant.

5.2. Evaluation metrics

The following metrics are used to evaluate the performance of the algorithms:

n	L	FH_k	FL_{ij}	Hub capacity	t_k
10	30	10^6	$10^5 \times C_{ij}$	Unlimited	10
54	60	10^7	$10^6 \times C_{ij}$	Unlimited	10

Table 2 **Input parameters for problem instances**

1. Number of Pareto optimal solutions (Size)
2. Hyper Volume indicator (HV)
3. Generational Distance (GD)
4. Inverted Generational Distance (IGD)
5. Spacing

The formula for HV is given in Guerreiro, Fonseca, and Paquete (2021). Given a point set $S \subset \mathbb{R}^d$ and a reference point $r \in \mathbb{R}^d$, the hypervolume indicator of S is the measure of the region weakly dominated by S and bounded above by r, i.e.

$$HV(S) = \Lambda(\{q \in \mathbb{R}^d | \exists p \in S : p \leq q \textbf{ and } q \leq r\})$$

Where Λ denotes the Lebesgue measure. Alternatively, HV can be defined as union of boxes:

$$HV(S) = \Lambda\left(\bigcup_{p \in S \wedge p \leq r} [p, r]\right)$$

where $[p,r] = \{x \in \mathbb{R}^d | p \leq x \leq r\}$ is the box delimited below by $p \in S$ and above by r

Denote $R = \{r_1, r_2, \ldots, r_{|R|}\}$ the reference point set (the true Pareto front); and $A = \{a_1, a_2, \ldots, a_{|A|}\}$ the set of non-dominated points found by an algorithm.

The Generational Distance (GD) (Ishibuchi et al. 2015) is defined as the average distance from each point in the approximation set to the nearest point in the reference set, i.e.

$$\text{GD}(A, R) = \frac{1}{|A|}\left(\sum_{i=1}^{|A|} d_i^p\right)^{1/p}$$

We choose $p = 2$, so d_i becomes the Euclidian distance from a_i to its nearest point in the reference set. $d_i = \min_{j=1}^{|R|} \|a_i - r_j\|_2$

The Inverted Generational Distance (IGD) (Ishibuchi et al. 2015) inverts the distances in GD and measures the average distance from any point in R to the closest point in A.

$$\text{IGD}(A, R) = \frac{1}{|R|}\left(\sum_{i=1}^{|R|} \hat{d}_i^p\right)^{1/p}$$

Metrics	Requires reference set R	Requires reference point r	Goal	Time complexity
Size	No	No	Maximize	$O(1)$
HV	No	Yes	Maximize	$O(n^{d-1})$ (While et al. 2006)
GD	Yes	No	Minimize	$O(n^2)$
IGD	Yes	No	Minimize	$O(n^2)$
Spacing	No	No	Minimize	$O(n^2)$

Table 3 An overview of used evaluation metrics

in which $\hat{d}_i$ is distance from r_i to its closest point in A: $\hat{d}_i = \min_{j=1}^{|A|} \|r_i - a_j\|_2$

Given the same R, $\mathrm{GD}(A,R) < \mathrm{GD}(B,R)$ means that solution set A has better convergence to R, compared to set B. Besides convergence, IGD also accounts for coverage of reference points. "If no solution associated with a reference point is found, the IGD metric value for the set will be large".

Spacing metric is calculated as the standard deviation of nearest neighbor distances in the solution set A. It measures the uniformity of the distribution of solutions in A. Lower value indicates better distribution.

An overview of evaluation metrics used is provided in Table 3. Some metrics (GD and IGD) require a reference set R to be defined. Since the true Pareto-front is unknown beforehand and impossible to obtain for large, complex real-world problem instances, we need a work-around for this. For each instance (each value of n), the reference set R_n is approximated by taking the non-dominated (best) solutions in the combined sets of solutions found by both algorithms in all 20 runs.

5.3. Results

NSMA's metrics have lower standard deviations than NSGA's, indicating more consistent performance across runs. Additionally, NSMA has higher HV than NSGA-II on average. NSMA generates more non-dominated points, distributes them more evenly, and has better convergence towards the *optimal* Pareto front. Notably, the GD metrics of NSMA is under 50% of that of NSGA-II across problem instances. The results demonstrate the efficiency of Local Search in improving the searching capability of the algorithm. This comes at a cost of higher complexity and increased computational time due to the additional Local Search phase.

The Pareto front solution is visualized in Figure 12.

Result for Complete Hub network (Uncapacitated Single Allocation HLP)

n	size		HV		Spacing (10^{-2})		GD (10^{-3})		IGD (10^{-4})	
	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA
10	17.4	19.1	0.801	0.804	16	16	1.41	0.805	156	109
54	271	343	0.927	0.931	2.45	2.04	1.64	0.729	9.94	7.91

Result for Incomplete Hub network variant

n	size		HV		Spacing (10^{-2})		GD (10^{-3})		IGD (10^{-4})	
	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA	NSGA	NSMA
10	5.5	7.55	0	0	0.159	0.154	7.75	12.4	682	389
54	431	464	0.945	0.955	2.07	1.68	1.16	0.452	7.35	5.39

Table 4 **Evaluation metrics of NSGA-II and NSMA on two problem variants**

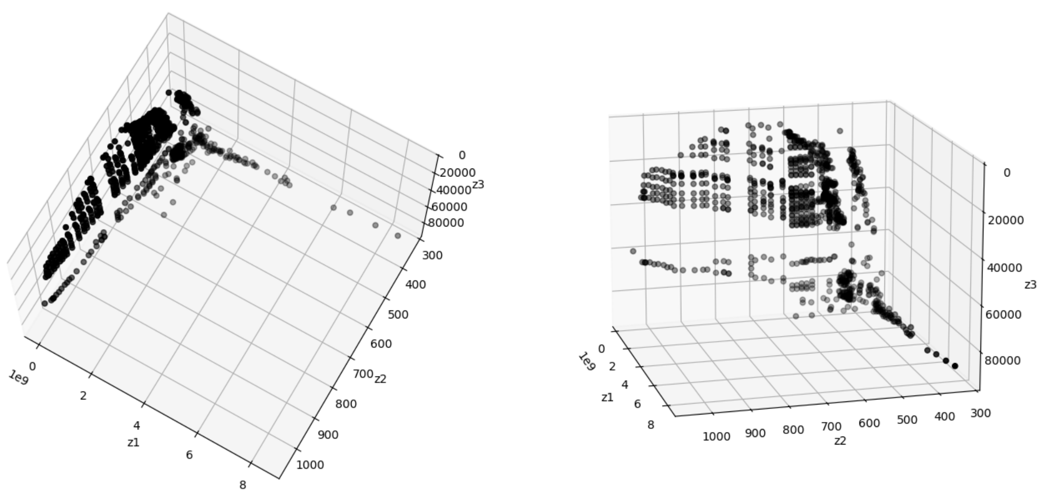

Figure 12 **Pareto set approximated by NSMA for Incomplete variant,** $n = 54$

ID	Z_1	Z_2	Z_3	Opened hubs
1	6.26×10^7	1.01×10^3	2.55×10^4	15
2	1.84×10^8	9.07×10^2	4×10^4	12 23
3	4.62×10^8	6.59×10^2	7.67×10^4	11 25 33
4	7.35×10^8	4.85×10^2	8.34×10^4	11 25 33 43
5	1.34×10^9	6.07×10^2	9.05×10^4	11 24 27 35 44 48 52
6	2.76×10^9	3.60×10^2	9.05×10^4	10 12 14 27 35 45 47 52 53

Table 5 **Some Pareto solutions found by NSMA for Incomplete variant**

The trade-off between objective values can be seen from the Table 5. As more costs being invested to open hub facilities and building hub network (Z_1), the service quality (Z_2) and customer's willingness to use the service (Z_3) increases. Decision makers can make choices based on their preferences.

6. Conclusion

In this study, we addressed the problem in rail transport infrastructure planning using a multi-objective hub location and network design model, with practical considerations for congestion at train station and incomplete network topology. An efficient hybrid memetic algorithm is proposed, showing superiority over NSGA-II baseline in terms of solution quality, population diversity and convergence speed. The algorithm is applied to a real-world dataset of 54 train stations along the Northern Line in Thailand, providing valuable insights for decision-makers in optimizing railway network design.

Future work will focus on the integration of hub capacity constraints into the model. Specifically, multiple levels of hub capacities (small, medium, large) will be introduced as decision variables to enhance the model's applicability. The algorithm will be refined to give useful decisions on routing of flows and size of hub facilities. We aim to improve the efficiency of the algorithms, as well as its applicability across different problem settings.

Notes

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